Architecture and Preprocessing Choices under Dataset Shift in Single-Channel EEG Sleep Staging: A Controlled Evaluation on Sleep-EDF and SHHS1

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Abstract

Background. Improvements in sleep staging on one dataset may not carry over to another. We examined how model and preprocessing choices affect accuracy, computational cost and cross-cohort errors.

Methods. Six configurations were compared on 78 Sleep-EDF subjects using shared 10-fold partitions. Comparisons covered sequence-model and training-recipe replacement, a bundled encoder/context replacement, and preprocessing. Paired subject bootstrap and Wilcoxon–Holm analysis assessed uncertainty. Transfer used 180 locked SHHS1 participants without weight updates. Record-wise normalisation used each complete unlabelled target recording and was therefore transductive.

Results. The fixed-scale ResNet-1D–TCN configuration (E3) reached Sleep-EDF macro-F1 0.7904 in the primary campaign, exceeding its record-wise-normalised counterpart (E6) by 0.0213 ($\mathrm{CI}_{95\%}$ [0.0122, 0.0307]; Holm $p = 0.0012$). A second seed preserved the direction but not Holm significance. E3 delivered a 3.76× forward-pass speed-up over the CNN–BiLSTM control, with 4.37× as many parameters and 28.4% more peak memory. On SHHS1, it improved subject-mean macro-F1 over the control by 0.0412 ($\mathrm{CI}_{95\%}$ [0.0314, 0.0512]). Yet both pipelines misclassified over 72% of N3 epochs as N2; N3 recall was 0.2610 for the control, 0.2582 for E3 and 0.2005 for E6.

Conclusions. The evaluated pipeline combined faster computation with improved aggregate transfer performance, but left a shared N3 deficit unresolved. Preprocessing choices and stage-specific errors are consequential alongside model design when assessing cross-cohort use.

Keywords: sleep staging; single-channel EEG; temporal convolutional network; preprocessing; domain shift; paired evaluation.

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1. Introduction

Sleep staging assigns one of five labels—W, N1, N2, N3 and REM—to every 30 s epoch of a polysomnographic recording. Manual scoring is slow, requires trained expertise, and retains irreducible disagreement between scorers; in the inter-scorer reliability programme of the American Academy of Sleep Medicine, N1 shows the lowest agreement of all stages [1]. An automated system therefore has to be judged not by overall accuracy alone, which is dominated by the most frequent stages, but by class-balanced metrics, by stability across individuals, and by its behaviour when the recording device, montage or population changes.

Progress in this field is difficult to read from published tables. Studies differ simultaneously in dataset version, subject partitioning, wake-trimming policy, filtering, normalisation, sequence length and model-selection rule, so a reported improvement cannot be attributed to the architecture that the paper nominally proposes. For practical development, the relevant questions are which pipeline choices improve prediction, what they cost computationally, and whether their benefits persist on another cohort. Answering these questions requires shared evaluation partitions and an explicit account of what changes between configurations, including their training procedures.

We take ZleepAnlystNet [2], a two-stage single-channel model that couples a bank of class-specific convolutional encoders to a BiLSTM, as an internal control. E1 replaces the BiLSTM configuration with a temporal convolutional network (TCN) [3], reusing the encoder and cached features but changing the sequence-model training recipe. E2 replaces E1's 75-dimensional current/previous/next (C/P/N) feature-context package with a 128-dimensional current-epoch residual encoder [4], also changing encoder training and selection. These are comparisons of specified configurations, not isolated architectural effects. Preprocessing contrasts then examine filtering and amplitude handling within the same ResNet-1D–TCN model and training configuration.

The study addresses three questions. (i) What predictive and computational benefits do the evaluated sequence-model and encoder/context replacements provide under shared subject-wise partitions? (ii) Which pipeline choices retain their benefits when transferred without weight updates to a separate cohort with a different EEG derivation? (iii) Which class confusions recur across model families, and how are they distributed between reference-label boundaries and stable sleep regions?

Our contributions are the following.

(1) A 10-fold subject-wise evaluation that compares six configurations on shared partitions, documents their model and training differences, and pairs aggregate effects with subject-level inference.
(2) A locked cross-cohort evaluation on 180 SHHS Visit-1 participants, showing that the ranking and magnitude of design choices can differ between internal validation and cross-cohort use.
(3) A class- and confusion-channel analysis showing that under-detection of N3 as N2 recurs in both the 15-CNN–BiLSTM control and ResNet-1D–TCN, while over-detection of REM from N2 is a second prominent E3 error channel. The evaluated record-wise normalisation pipeline does not resolve N3 under-detection.
(4) A quantified operational trade-off: the implemented ResNet-1D–TCN pipeline reduces measured forward latency and training-and-validation wall-clock time relative to the 15-branch CNN–BiLSTM control, while increasing parameter count and peak memory. These operational gains are distinguished from predictive differences within and across cohorts.

Throughout, we distinguish pre-specified comparisons, secondary evidence and post-hoc contrasts, and we confine claims to the evidence class that supports them.

2. Related work

DeepSleepNet combined a dual-branch CNN with a BiLSTM to capture epoch morphology and temporal dependence [5]. AttnSleep introduced attention over multi-resolution features [6], and SleepTransformer applied self-attention with explicit uncertainty quantification [7]. Together these establish that sequential context matters substantially. They also illustrate the comparability problem: differences in dataset version, subject split and preprocessing between these works are large enough that cross-table comparison cannot substitute for an ablation carried out inside one protocol, a point emphasised in recent reviews of the field [8].

Single-channel systems also span materially different input representations. SleepInceptionNet systematically compared raw, spectral and time–frequency inputs and selected a continuous-wavelet, central-channel Inception model that scores one epoch at a time [9]. That study reinforces two points relevant here: a single EEG channel can support five-stage scoring, and preprocessing choice is part of the model definition rather than a neutral implementation detail.

Cross-dataset sleep staging is already an active methodological problem. For example, He et al. [10] trained a domain-adaptation network across Sleep-EDF, SHHS1 and the PhysioNet Challenge data. ADAST instead uses domain-specific attention, adversarial alignment and iterative pseudo-label self-training on unlabelled target data [11]. These approaches adapt model representations or decision functions using the target domain; our label-free E6 condition only estimates recording-level normalisation statistics, with no learned target-domain or weight adaptation. Van Der Donckt et al. [12] further showed that increasingly complex deep architectures need not outperform a carefully designed conventional pipeline under a matched protocol. Our contribution is a joint assessment of pipeline substitutions, preprocessing, computational cost and cross-cohort error structure. The paired comparisons quantify benefits within a common protocol; the cross-model analysis tests whether an aggregate transfer gain also addresses the dominant stage-specific failure. Neither question is answered by comparing scores from unrelated benchmark tables.

Sleep-EDF Expanded is the standard public benchmark for single-channel EEG [13, 14]. The Sleep Heart Health Study (SHHS) is a large multi-centre cohort assembled to study sleep-disordered breathing and cardiovascular risk [15], distributed through the National Sleep Research Resource (NSRR) [16]. The two differ in age, comorbidity, recording hardware, montage and label distribution. SHHS therefore tests a form of cohort change not represented by a held-out split of Sleep-EDF. We use it to assess cross-cohort robustness, without equating retrospective transfer testing with clinical deployment validation.

Two further strands of work inform the interpretation. Slow oscillations have a spatially varying amplitude and propagate across the cortex [17], while deep-sleep scoring uses amplitude criteria [18, 19]. A recent SHHS analysis examined how those criteria relate to age- and sex-associated differences in N3 scoring [20]. Such work provides context for transfer between derivations and populations, rather than a mechanism established by the present experiment. Differences in class prevalence can also affect predictions; prior correction and probability calibration address related but distinct problems [21, 22]. We therefore examine class-wise errors and label proportions without treating marginal prediction counts as a calibration measure.

3. Materials and methods

3.1. Data and labels

The Sleep Cassette subset of Sleep-EDF Expanded comprises 153 recordings from 78 subjects; we use the Fpz–Cz derivation at 100 Hz with 30 s epochs (3,000 samples per epoch). Annotations are mapped to W, N1, N2, N3 and REM, merging the original stages 3 and 4 into N3; Movement and Unknown epochs retain their temporal position but are excluded from the loss and from all metrics. After windowing, 195,767 epochs remain, of which 195,469 are scored and 298 are Movement/Unknown.

An evaluation window is applied identically to both cohorts. It is anchored on the first and last true sleep epoch (any of N1, N2, N3, REM) and extended by 30 min of wake on each side, i.e. 60 epochs per side. This policy is stated explicitly because it strongly affects the wake fraction and hence every aggregate metric, and because it is one of the principal reasons published numbers are not directly comparable. It is a benchmark window, not a deployable inference rule, since its anchors use ground-truth labels.

The SHHS Visit-1 selection used the metadata quality field `overall_shhs1`, admitting values 3–7. Eligible participant identifiers were ranked deterministically by SHA-256 with selection seed 42, then independently ranked for role assignment: 5 adaptation, 15 validation, 180 test and 20 reserve participants. No reserve replacements were required. The 180-person test sample was locked before prediction review; the adaptation and validation participants were not used to update model weights. The EEG channel is C4–A1 at 125 Hz, resampled to 100 Hz with a polyphase anti-aliasing filter (up 4, down 5, Kaiser window $\beta = 5.0$) applied to the continuous signal before epoching. SHHS labels are mapped $0 \to$W, $1 \to$N1, $2 \to$N2, $\{3, 4\} \to$N3, $5 \to$REM, $\{6, 9\} \to$excluded. Table 1 reports both label distributions.

The prior shift between cohorts is substantial: total variation distance 0.187, $D_{\mathrm{KL}}(\mathrm{EDF} \,\|\, \mathrm{SHHS}) = 0.101$ nats. SHHS1 contains proportionally 2.02× as much N3 and 0.38× as much N1. We report the per-class ratios in Table 1 because Section 4.5 uses their *direction* as evidence.

Table 1. Scored-label distribution in the two evaluation samples.

Stage	Sleep-EDF		SHHS1 test		
	Epochs	Share	Epochs	Share	Ratio to EDF
W	65,941	33.73%	36,983	21.88%	0.65
N1	21,522	11.01%	7,002	4.14%	0.38
N2	69,132	35.37%	76,287	45.14%	1.28
N3	13,039	6.67%	22,806	13.49%	2.02
REM	25,835	13.22%	25,934	15.34%	1.16
Total scored	**195,469**	**100%**	**169,012**	**100%**	

3.2. Preprocessing conditions

To evaluate the individual and joint impacts of architectural design and preprocessing choices, we propose an end-to-end processing pipeline, as illustrated in Figure 1. The signal is first calibrated and processed continuous-wise to preserve amplitude relationships, after which a 1D-ResNet epoch encoder and a TCN sequence model extract local and temporal characteristics, respectively.

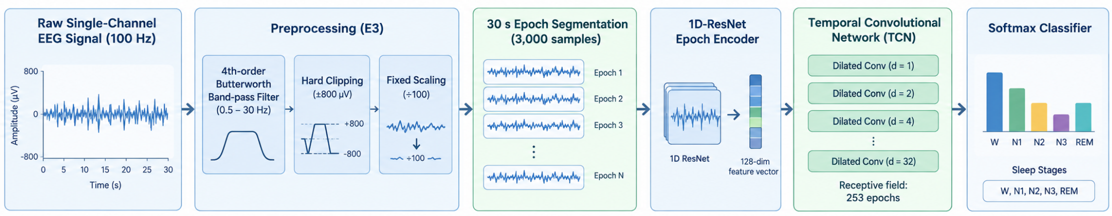


Figure 1. End-to-end processing pipeline for the proposed ResNet-1D–TCN single-channel sleep staging framework (E3 configuration). The raw continuous EEG signal is sequentially preprocessed (band-pass filtered, clipped, and scaled), segmented into 30 s epochs (3,000 samples), encoded into 128-dimensional feature vectors by a 1D-ResNet, and temporally modeled by a six-block non-causal TCN with a 253-epoch receptive field before final Softmax classification.

All conditions operate on the continuous calibrated signal in microvolts before epoching. Four active signal conditions enter the performance analysis, and one audit-only condition (E5) records a planned clipping ablation:

- **Raw** (E0–E2): no filtering, no clipping, no rescaling; native μV retained.
- **Band-pass only** (E4): zero-phase Butterworth band-pass, order 4, 0.5–30 Hz, implemented as second-order sections with forward–backward filtering.
- **Band-pass plus clipping** (E5; audit only): the E4 signal followed by a hard clip at ±800 μV. Across all 153 Sleep-EDF recordings, E5 and E4 were bitwise identical and the measured clipped-sample fraction was zero. Fold 0 was retained as an audit artifact; the remaining folds and all performance tests were cancelled before scores were inspected because E5 did not define an independent input condition.
- **Fixed-scale amplitude handling** (E3): the same band-pass, followed by a hard clip at ±800 μV and division by a constant factor of 100. The clip threshold is an absolute constant identical for every recording, so no record-level or cohort-level statistic is estimated and cross-subject leakage is not introduced; relative amplitude between recordings is preserved.
- **Per-record z-score** (E6): the same band-pass and clip, followed by subtraction of the mean and division by the standard deviation computed over the whole filtered, clipped recording. This uses no target labels but does use all samples of the target recording, removes absolute amplitude differences between

Table 2. Experimental configuration codes. The identifiers denote signal–model combinations, not a performance ranking or a sequence of model versions.

ID	Signal condition	Epoch encoder	Sequence model and role
E0	Raw	15-branch CNN	BiLSTM; internal control
E1	Raw	15-branch CNN	TCN; sequence-model and training-recipe replacement
E2	Raw	ResNet-1D	TCN; encoder/context and encoder-training replacement
E3	Band-pass + clip + fixed scale	ResNet-1D	TCN; fixed-scale configuration
E4	Band-pass only	ResNet-1D	TCN; band-pass configuration
E5	Band-pass + clip	ResNet-1D	TCN; audit only, identical input to E4
E6	Band-pass + clip + z-score	ResNet-1D	TCN; alternative rescaling

recordings and is transductive at the recording level rather than purely inductive.

No notch filter is applied: 50 Hz lies outside the 0.5–30 Hz pass-band and at the Nyquist frequency of the 100 Hz signal. The contrast E3 – E6 therefore evaluates a practical preprocessing choice: preserving calibrated cross-recording amplitude relationships or applying record-wise normalisation. Because clipping never activated in the Sleep-EDF E5 audit, E3 and E4 differ on Sleep-EDF by the fixed division by 100; this identity is used only to interpret the observed contrast and does not establish equivalence or a hardware-independent scaling rule.

3.3. Model configurations

Six active configurations (E0–E4 and E6) enter performance analysis; E5 documents the cancelled clipping-only contrast. The control encoder concatenates five-class softmax outputs from 15 CNN branches spanning current, previous and next epochs (C/P/N), giving 75 features per central epoch. Context stays within each recording, with edge-epoch replication. ResNet-1D produces 128 features from the current epoch only. The non-causal six-block TCN has a 253-epoch receptive field; the BiLSTM control has 128 hidden units per direction. Both sequence models process the retained recording with masked padding. These are offline models: the BiLSTM and TCN use future context, and zero-phase filtering and E6 normalisation use future signal samples. The forward benchmark is not a measure of online decision latency. Layer dimensions and all hyperparameters are reported in Supplementary Methods S1.

3.4. Two-stage training and subject-wise cross-validation

The 78 subjects are partitioned into 10 subject-disjoint folds. In each evaluation round one fold is the test fold, the next fold (modulo 10) is the validation fold and the remainder are training folds. Both nights of a subject always share the same role. The same partition is used for every configuration, and the test fold never participates in model selection. The training API accepts only training and validation loaders, and encoder fitting and feature caching follow the same subject-fold assignments.

Training has two stages. The epoch encoder uses class weights calculated from training subjects and is selected on validation data: minimum weighted validation loss for each control CNN, maximum validation macro-F1 for ResNet. Its selected checkpoint is hash-verified and used to cache features without gradients. The sequence model is fitted on the frozen cache using masked cross-entropy and selected by validation macro-F1. E1 reuses E0 encoder checkpoints but changes the sequence optimiser settings: learning rate 0.01 versus 0.0005, batch size 4 versus 8 recordings, maximum 1,000 versus 300 epochs and patience 10 versus 30 for E0 and E1, respectively. E2 additionally changes encoder optimisation and selection alongside the feature/context replacement. Supplementary Methods S1 provides the complete recipes. The comparisons therefore preserve subject partitions and evaluation, not every training choice.

3.5. Test locking, statistical analysis and effect sizes

All six configurations completed validation-based selection before the test set was opened, once. In Sleep-EDF, pooled macro-F1 is the descriptive headline metric, but the unit of statistical inference is the subject. Paired subject-cluster bootstrap with 10,000 resamples estimates confidence intervals for the difference in pooled macro-F1, while a two-sided Wilcoxon signed-rank test [23] assesses the subject-level distribution of per-subject differences [24]. Holm correction [25] is applied over the four pre-specified comparisons E1 – E0, E2 – E1, E3 – E2 and E3 – E6. Pooled macro-F1 is computed from the sum of subject confusion matrices; subject-mean macro-F1 is the arithmetic mean of individual subjects' scores. They are not interchangeable, and pooled macro-F1 is not a weighted average of individual macro-F1 values. The bootstrap interval describes the pooled-score difference, whereas Wilcoxon evaluates the paired subject differences through their signed ranks. All bootstrap intervals are marginal 95% intervals, not multiplicity-adjusted simultaneous intervals. Metrics use five fixed classes and assign zero when a class-wise denominator is zero. E3 – E0 is labelled post-hoc throughout.

We also report win/tie/loss counts, sign dominance $(W - L)/(W + L)$ and an unadjusted exact sign test as descriptive robustness checks; they are not part of the pre-specified inferential family.

The completed seed-123 campaign repeated the evaluation with a second fixed training seed on the same partition. The two seeds are analysed separately; p-values are never pooled, and two fixed seeds do not characterise the distribution over initialisations. We report this repeat as sensitivity evidence, not an additional confirmatory campaign. The archived configuration lists seeds 42, 123 and 2025; only completed campaigns 42 and 123 contribute results here.

For SHHS, each configuration uses all ten Sleep-EDF outer-fold checkpoints. Softmax probabilities are averaged in float64 across folds before the final argmax for each epoch. The pre-locked primary criterion is subject-mean macro-F1, with E3 – E0 and E3 – E6 as the two locked comparisons under the same subject-cluster bootstrap and Wilcoxon–Holm treatment, now with bootstrap intervals for the subject-mean difference. The E1/E2 analysis is secondary evidence on an already-opened sample, and E3 – E2 is explicitly post-hoc. The seed-123 E4 extension has its own four-comparison Holm family, reported in the supplement. Reference-label boundary and stable-region analyses are post-hoc offline diagnostics; their exact masks, exclusions and supports are defined in Supplementary Methods S1.

3.6. Operational evaluation

Inference timing was measured on identical hardware (Tesla V100) for all configurations and excludes I/O, preprocessing and training. Each measurement used a batch of one synthetic 100-epoch sequence, 20 warm-up passes and three rounds of 100 timed forward passes. Configuration order was shuffled in each round; we report the median over all 300 passes. This benchmark measures the implemented full forward pipeline for one fixed tensor shape, not overnight wall-clock inference or algorithmic complexity.

4. Results

4.1. Out-of-fold performance on Sleep-EDF

E3 leads the primary seed-42 campaign on all four metrics (Table 3). Table 4 distinguishes this aggregate ranking from the evidence for paired improvements across subjects.

E3 – E6 is the only pre-specified contrast with both a positive pooled-metric bootstrap interval and a Holm-significant subject-level Wilcoxon result. The E1 – E0 and E2 – E1 intervals include zero. E3 – E2 has a positive pooled interval but a non-significant Wilcoxon result.

E3 – E2 deserves particular attention because it is the clearest case of estimand disagreement in the study. Its pooled bootstrap interval marginally excludes zero ($[0.000305, 0.014520]$), yet only 37 of 78 subjects improve, $\Delta_{\text{sign}} = -0.051$, and both the Wilcoxon and sign tests are far from significance ($p_{\text{sign}} = 0.734$). The analyses capture different aspects of performance: an aggregate gain need not be accompanied by improvement for most subjects. Their disagreement does not, by itself, identify which subjects drive the pooled change. The evidence supports a small pooled improvement for this contrast, but not a consistent subject-level advantage.

Table 3. Out-of-fold test performance on Sleep-EDF (seed 42, pooled over all folds).

Configuration	macro-F1	**Accuracy**	κ	**F1 (N1)**
E0	0.7754	0.8262	0.7614	0.5009
E1	0.7802	0.8315	0.7683	0.5115
E2	0.7835	0.8341	0.7716	0.5180
E3	**0.7904**	**0.8363**	**0.7754**	**0.5276**
E4	0.7891	0.8360	0.7746	0.5270
E6	0.7691	0.8230	0.7573	0.5124

Table 4. Pre-specified paired comparisons on Sleep-EDF macro-F1 (n = 78 subjects). Holm correction is over the four pre-specified rows; sign-test statistics are unadjusted supplementary robustness checks. E3 – E0 is post-hoc and is excluded from the Holm family.

Comparison	Δmacro-F1	$\mathrm{CI}_{95\%}$	p_{Holm}	**W/T/L**	Δ_{sign}	p_{sign}
E1 – E0	0.004811	[−0.000915, 0.010193]	0.1022	51/0/27	+0.308	0.0088
E2 – E1	0.003251	[−0.002370, 0.008808]	0.1036	44/0/34	+0.128	0.3082
E3 – E2	0.006962	[0.000305, 0.014520]	0.8989	37/0/41	−0.051	0.7343
E3 – E6	**0.021319**	**[0.012179, 0.030698]**	**0.0012**	49/0/29	**+0.256**	**0.0308**
E3 – E0 (post-hoc)	0.015024	[0.005746, 0.025871]	—	49/0/29	+0.256	0.0308

For the post-hoc E3 – E0 contrast, the complete E3 pipeline adds 0.0150 macro-F1 over E0, with a fully positive interval. This comparison represents the combined architecture, preprocessing and training configuration rather than any single component.

4.2. Per-class and seed-sensitivity checks

Detailed per-class metrics, the Sleep-EDF confusion matrix and both seed-sensitivity tables are provided in Supplementary Tables S4–S7. In domain, N1 is E3's hardest class and the largest E3–E6 F1 difference is on N3 (+0.0628). All four pre-specified differences remain positive under seed 123, but only E3 – E6 has a fully positive bootstrap interval in both runs; its Holm-adjusted result is significant for seed 42 only and its magnitude falls from 0.0213 to 0.0102. The direction is therefore stable across the two fixed seeds, whereas the inferential strength is not.

4.3. Complexity, latency and training cost

As Table 5 shows, the ResNet-1D–TCN family provides a measured 3.76× speed-up over the CNN–BiLSTM control while carrying 4.37× the parameters and 28.4% more peak allocated memory. We quantify this operational trade-off between predictive power and computational cost across both evaluation settings in Figure 2. As visualized in the cross-cohort panel, the ResNet-1D–TCN configuration (E3) translates its computational efficiency into robust generalization on the unseen SHHS1 cohort (Section 4.4), rather than trading accuracy away for speed. The implementation is therefore faster but not smaller. In the forward benchmark, E1 reuses the same encoder as E0 and changes the sequence model, giving a 1.07× speed-up; the larger E2–E0 difference is associated primarily with replacing 15 independently executed CNN branches by one ResNet-1D encoder, together with the full-pipeline implementation.

Table 5. Full-pipeline forward benchmark on a Tesla V100 (fold 0, seed 42; batch 1, sequence length 100). Throughput is in epochs s^{-1} and peak allocated memory in MiB; I/O and preprocessing are excluded.

ID	Parameters	Median (ms)	Throughput	Peak mem.	Speed-up
E0	248,630	13.4315	7,445	58.81	1.00×
E1	640,950	12.5111	7,993	60.31	1.07×
E2	1,085,578	3.5750	27,972	75.54	3.76×
E3	1,085,578	3.5687	28,021	75.54	3.76×
E6	1,085,578	3.5756	27,967	75.54	3.76×

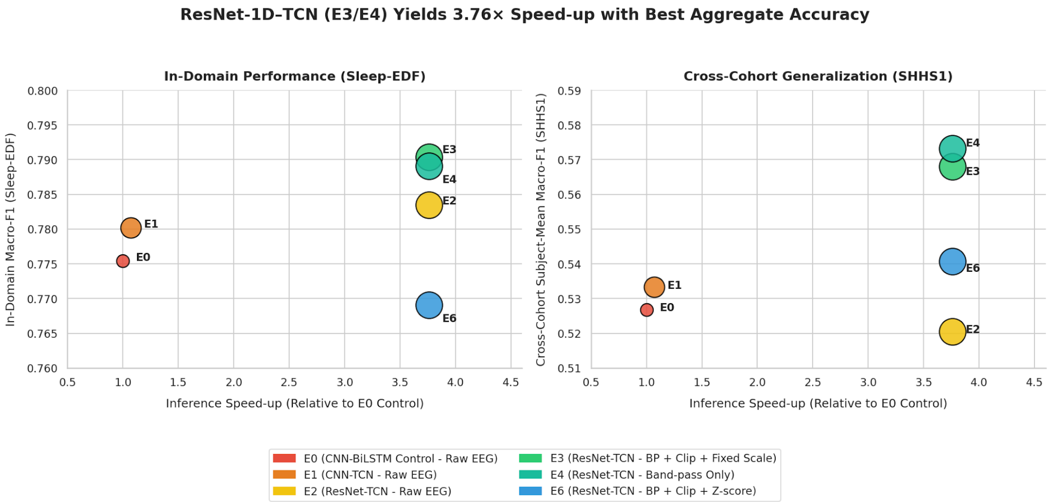
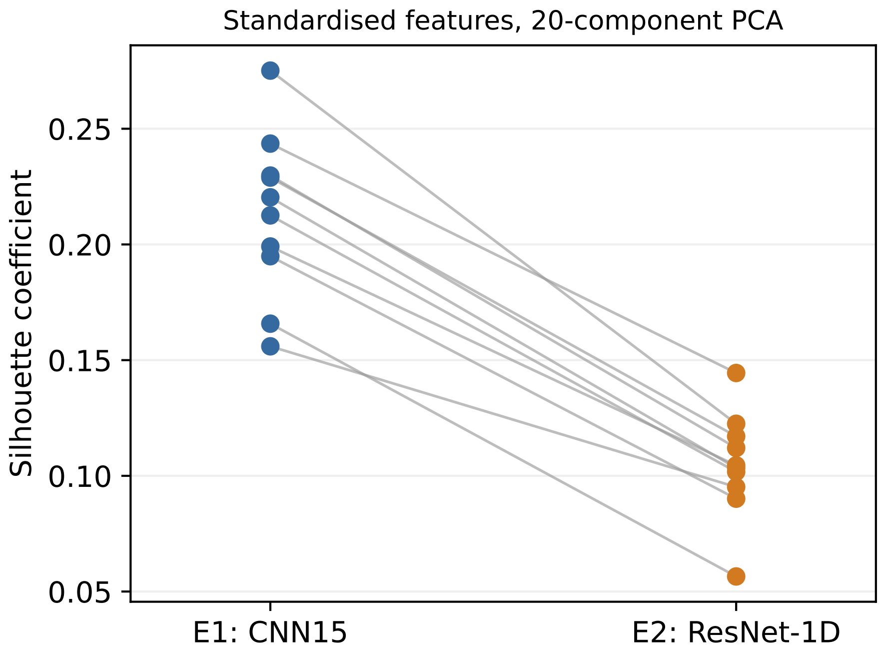


Figure 2. Inference speed-up versus macro-F1 **trade-off across Sleep-EDF (in-domain) and SHHS1 (cross-cohort) datasets.** The proposed ResNet-1D–TCN configuration (E3) achieves a 3.76× forward-pass speed-up over the CNN–BiLSTM control (E0) while simultaneously improving aggregate classification performance in both evaluation settings.

In the sequential seed-123 campaign, complete training and validation for E2 and E3 was observed to be 12.2× and 10.2× faster than E0. E1 reused E0 encoder checkpoints and therefore has no end-to-end runtime comparison. Full timings for the specified training and early-stopping recipes are reported in Supplementary Table S8.



Supplementary Figure 1. Silhouette coefficient of standardised features across 20-component PCA. The figure provides a visual comparison of the spatial representation structure between configuration E1 (15-branch CNN) and E2 (ResNet-1D). Although the ResNet-1D encoder (E2) exhibits lower cluster density (lower Silhouette coefficient) in the PCA-reduced feature space, its single-branch, current-epoch architecture drastically reduces complexity and optimises computational speed when coupled with the TCN sequence model (as quantified in Table 5). This illustrates the trade-off between feature geometry compactness and end-to-end pipeline efficiency.

Additional descriptive representation and context-group analyses are reported in Supplementary Fig. 1 and Table S17; they provide complementary evidence on feature geometry and temporal context.

4.4. Cross-cohort evaluation on SHHS1

The seed-123 campaign preserves the direction and Holm significance of both SHHS contrasts, whereas the in-domain E3 – E6 comparison no longer reaches the Holm threshold. This sensitivity result uses the same SHHS sample; its four-comparison extension family is reported separately in the supplement.

Both locked comparisons support E3 (Table 7), with improvements in 138 and 125 of 180 subjects, respectively. Sign dominance is +0.533 and +0.389; the unadjusted exact sign tests give $p = 3.8 \times 10^{-13}$ and 1.9×10^{-7}. Supporting metrics move together: relative to E0, E3 gains 0.0338 pooled macro-F1, 0.0368 accuracy and 0.0461 κ. Within ±1 epoch of a transition, E3 gains 0.0366 over E0 and 0.0183 over E6.

The secondary configuration analysis gives a different picture. E1 – E0 remains positive but small (+0.0065 subject-mean macro-F1). E2 – E1 is *negative* out of domain (−0.0128, interval entirely below zero), so the bundled replacement of E1's 75-dimensional C/P/N package by the current-epoch 128-dimensional ResNet-1D representation does not generalise better on unprocessed SHHS signals. Because the entire feature/context package changes, the reversal cannot be attributed to the encoder alone. The post-hoc E3 – E2 contrast is the largest observed configuration difference (+0.0475, $\mathrm{CI}_{95\%}$ [0.0372, 0.0579], $\Delta_{\mathrm{sign}} = +0.633$, 147/180 subjects improved). E3 and E2 use the same model and specified training recipe with different preprocessing. This observed contrast is larger than either the sequence-model/recipe or encoder/context contrast on this sample.

A seed-123 extension evaluating E4 (band-pass only) provides a more specific secondary contrast. E4 reached subject-mean macro-F1 0.5732, pooled macro-F1 0.6147, accuracy 0.7064, $\kappa = 0.5869$ and N1 F1 0.3492, exceeding same-seed E2 by 0.0323 and E3 by 0.0098. The SHHS-specific preprocessing audit recorded zero clipped samples across the 200 processed adaptation, validation and test recordings. E4 and

Table 6. Cross-cohort performance on 180 locked SHHS1 participants (seed 42). All model weights were trained on Sleep-EDF. E6 additionally uses complete-record, unlabelled target statistics. Intervals are subject-cluster bootstrap intervals for subject-mean macro-F1.

ID	**Subject-mean** macro-F1 ($CI_{95\%}$)		**Pooled** macro-F1	**Accuracy**	κ
E0	0.5268	[0.5101, 0.5431]	0.5761	0.6648	0.5339
E3	**0.5680**	**[0.5503, 0.5850]**	**0.6099**	**0.7016**	**0.5801**
E6	0.5407	[0.5227, 0.5584]	0.5732	0.6742	0.5408

Table 7. Locked paired comparisons on SHHS1 subject-mean macro-F1 (n = 180).

Comparison	Role	Δ	$CI_{95\%}$	p_{Holm}	W/T/L	Δ_{sign}
E3 – E0	locked primary	0.0412	[0.0314, 0.0512]	2.65×10^{-13}	138/0/42	+0.533
E3 – E6	locked primary	0.0274	[0.0182, 0.0370]	1.87×10^{-8}	125/0/55	+0.389

E3 therefore differ on these inputs by the fixed division by 100. The extension therefore associates band-pass filtering with most of the E4–E2 gain and provides no evidence that the additional fixed scaling improves transfer. These comparisons remain secondary because they were conducted after the SHHS sample had been opened.

4.5. Cross-model errors and their temporal distribution

E3's pooled macro-F1 is 0.7904 on Sleep-EDF and 0.6099 on SHHS1, a difference of 0.1805. Because Sleep-EDF uses out-of-fold predictions whereas SHHS uses ten-model ensembles, the 0.1805 difference is interpreted descriptively. We examine it by class and confusion channel to determine which errors remain despite E3's aggregate transfer advantage. Throughout this analysis, A→B denotes a reference-labelled A epoch predicted as B; it does not denote a physiological stage transition over time.

N3 is the dominant class loss: E3 N3 F1 falls from 0.8085 on Sleep-EDF to 0.4055 on SHHS1, with precision 0.9440 but recall 0.2582. E0 shows nearly the same recall (0.2610) and N3→N2 rate (72.3% versus 73.1% for E3), so E3's aggregate transfer advantage does not resolve N3 and the failure is not specific to the ResNet-1D–TCN configuration (Table 8). N2→REM is the second-largest E3 confusion channel.

An oracle calculation moves the counts in one selected off-diagonal cell of E3's SHHS confusion matrix to the correct diagonal cell, then recomputes macro-F1. Correcting N3→N2 raises macro-F1 to 0.7444; correcting N2→REM raises it to 0.6596. These changes correspond to 74.5% and 27.5% of the observed pooled-score difference. The calculation uses true labels, and its nonlinear effects are neither additive nor achievable-performance estimates.

The reference-label boundary diagnostic adds temporal detail. For E0 and E3, N3 recall near changes is 0.6121/0.6326 on Sleep-EDF but only 0.0633/0.0733 on SHHS1. Stable-region SHHS recall remains low at 0.4008/0.3900. Thus, N3 under-detection extends beyond the boundary neighbourhoods; it cannot be described solely as an error at label changes. The region definitions and supports are provided in the supplement, including the epochs omitted from the near-versus-stable comparison.

Furthermore, as detailed in the context ablation study (Figure 3), removing the explicit Current/Previous/Next (C/P/N) epoch context does not significantly degrade boundary performance (Holm p = 1.000). This strongly supports the premise that sequence models with extended receptive fields, such as the evaluated TCN (253 epochs), successfully capture broad temporal dynamics natively, rendering the localized C/P/N clustering redundant.

E6 provides a label-free but transductive sensitivity check. Its complete-target-record z-score lowers N3 recall to 0.2005; boundary-neighbourhood recall is 0.0721 versus 0.0733 for E3. The tested record-wise normalisation pipeline therefore does not remedy N3 under-detection. Detailed class-wise, confusion, oracle and transition results are in Supplementary Tables S12–S16.

Table 8. N3 under-detection across the primary SHHS1 configurations (seed 42). All models share 22,806 reference N3 epochs. The final column is the proportion of those epochs predicted as N2.

ID	N3 precision	N3 recall	N3 F1	N3→N2 count	Rate
E0	0.9007	0.2610	0.4048	16,480	72.3%
E3	0.9440	0.2582	0.4055	16,674	73.1%
E6	0.9052	0.2005	0.3283	17,764	77.9%

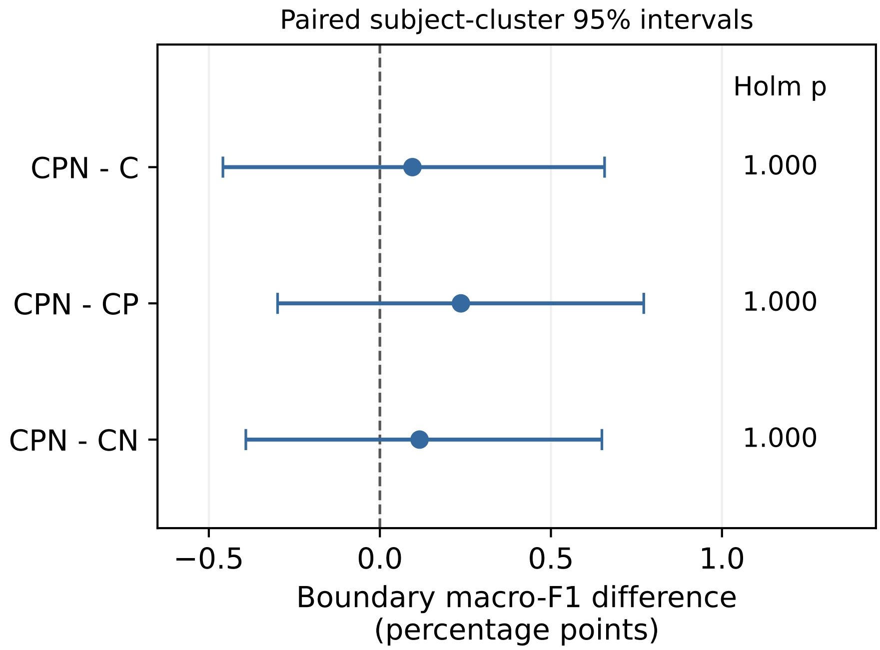


Figure 3. Context-ablation effects on boundary macro-F1.

5. Discussion

5.1. What the pipeline comparisons establish

The main result combines operational and cross-cohort gains at the pipeline level. ResNet-1D–TCN processes the benchmark input 3.76× faster than the CNN–BiLSTM control, while using more parameters and memory. Its observed training-and-validation time is also shorter, although the recipes and stopping rules differ. On the locked SHHS1 sample, E3 improves subject-mean macro-F1 over E0 by 0.0412, with a positive interval and gains in 138 of 180 participants. These results make the configuration a useful computational alternative with a demonstrated aggregate transfer benefit in this setting.

The intermediate comparisons explain why that benefit should not be credited simply to a newer encoder or sequence model. E1–E0 changes both the sequence model and its training recipe; E2–E1 changes the encoder, feature context and encoder training. Neither contrast reaches the subject-level Holm threshold on Sleep-EDF, and E2–E1 reverses direction on SHHS1. By contrast, the observed post-hoc E3–E2 preprocessing difference is larger on SHHS1. The seed-123 extension further shows that E4, which retains band-pass filtering without fixed scaling, outperforms both E2 and E3 on that sample. Signal handling is therefore a material part of the evaluated model's transfer behaviour, rather than a negligible implementation detail. These configuration-level comparisons do not isolate the contribution of each preprocessing operation.

5.2. Aggregate improvement leaves a shared N3 deficit

The class-wise analysis changes the interpretation of E3's overall advantage. E0 and E3 both misclassify more than 72% of true N3 as N2, with recall near 0.26 despite precision above 0.90. The alternative E6 pipeline has still lower N3 recall. Thus, choosing the better aggregate score does not address this common failure. The oracle calculation shows the numerical importance of N3→N2 for E3, while the regional analysis shows that the deficit is present in stable N3 as well as around reference-label changes. Together, these observations provide a more specific target for subsequent validation than the overall transfer score alone.

Several explanations remain plausible. The datasets use different EEG derivations, and population, acquisition and scoring conditions change at the same time. Evidence on slow-wave spatial structure and N3 scoring criteria provides reasons to investigate these factors [17, 20], but the present comparison cannot separate them. Nor does E6 settle the role of amplitude: record-wise z-scoring changes the representation used during source training as well as target inference. Its failure to remedy N3 under-detection does not rule out amplitude- or montage-related contributors.

The class-prior difference suggests another testable explanation. Under the assumptions required for prior correction, its direction would increase N3 and decrease N1 emissions; whether that improves classification remains to be measured. The observed prediction-to-reference ratios are marginal counts, not evidence of probability calibration. Calibration, limited fine-tuning and montage-aware representations are possible follow-ups, not remedies validated by this study. Any intervention should report N3 precision or false-positive rate alongside recall, because increasing N3 predictions alone can trade one error for another. N2→REM is a second candidate channel for E3.

5.3. Positioning and limits of generalisation

The contribution lies in connecting paired pipeline comparisons to their computational trade-offs and cross-cohort error structure. This complements studies of preprocessing and input representations, such as SleepInceptionNet [9], and systematic comparisons of traditional and deep pipelines [12]. It differs from adaptation-method studies because no target-domain weight update is evaluated. E0 is our reimplementation of the ZleepAnlystNet design under the stated subject-wise protocol, not a claim to reproduce every original training detail.

Published scores cannot be ranked directly against these results without matching dataset subset, subject splits, derivation, wake trimming and preprocessing. Wake trimming changes the evaluated class composition; its isolated numerical effect was not tested here. Likewise, the difference between Sleep-EDF out-of-fold predictions and SHHS ten-model ensembles means that the cross-cohort score difference is not a controlled estimate of any one source of domain shift.

The evidence covers one transfer direction and one locked SHHS1 sample. Two training seeds provide a sensitivity check, not a precise estimate of initialisation variability. The SHHS component extensions and reference-label diagnostics were evaluated after the cohort had been opened; they refine the interpretation of the primary results but are not independent confirmations. Subject-cluster inference preserves the dependence between recordings, while pooled stage metrics do not describe the full distribution of individual patients' errors. Additional subject-level summaries and an unseen target sample would strengthen that aspect of validation.

Finally, these are offline, retrospective evaluations. The sequence models use future context, and the benchmark window and regional masks use reference labels. Neither the forward speed-up nor the error localisation constitutes a real-time or clinical validation result. Independent validation of the N3 pattern, followed by a separately specified correction study if warranted, is a natural extension of the empirical findings.

6. Conclusion

A shared subject-wise protocol allowed six active configurations to be compared on the same Sleep-EDF folds; a seventh planned configuration, E5, was excluded before performance analysis because it was bitwise identical to E4. The sequence-model/recipe and encoder/context replacements did not establish a stable subject-level predictive advantage on Sleep-EDF after correction for multiple comparisons. ResNet-1D–TCN offers an operational alternative that is 3.76× faster in the controlled forward benchmark and 10.2–12.2× faster in observed training-and-validation wall-clock time, at the cost of more parameters and memory.

On SHHS1, the fixed-scale E3 procedure outperformed both locked references, while the largest observed configuration difference was associated with preprocessing under a fixed architecture and training recipe. These findings demonstrate the value of evaluating signal handling alongside model design, rather than assuming that an in-domain model ranking will carry over to another cohort.

The class-wise results reveal a limitation hidden by that aggregate improvement. E0 and E3 both systematically under-call N3 as N2, showing that E3's overall transfer advantage does not solve N3 detection; N2 is also frequently over-called as REM by E3. Per-record z-scoring does not repair the N3 deficit, and the cross-cohort diagnostic locates poor recall both near reference-label changes and within stable SHHS1 regions. The study therefore delivers both a computationally faster pipeline with improved aggregate transfer performance and a specific cross-model error pattern for further validation. Whether calibration, adaptation or montage-aware signal representations can correct that pattern remains an experimental question.

Reproducibility and data availability

Code, protocol snapshots, data partitions and aggregate analyses are retained in the project repository at `https://github.com/quanuiter/SleepTCN`. The repository does not contain restricted participant-level SHHS data, identifiers or predictions. Sleep-EDF Expanded is available through PhysioNet; SHHS access is managed by NSRR. Participant-level SHHS records and predictions are not included in the manuscript package. The historical snapshot `configs/shhs_zero_shot_v1.json` matches the protocol hash in the original SHHS run manifest. The primary campaign's 540 ensemble prediction files were checked against their recorded SHA-256 values, and their confusion counts reproduce the reported pooled results. This verifies those retained artifacts; it is not a regeneration of predictions or an end-to-end rerun. The expanded post-run protocol record is preserved separately. Further provenance details are given in the supplement and the repository artifact index.

The experimental revision is fixed at Git commit `f7c22e7`.

Ethics

This study is a secondary analysis of de-identified repository data; no new participants were recruited. The authors confirm that access to SHHS for this analysis was obtained through the National Sleep Research Resource and that no ethics approval was required for this secondary analysis of de-identified repository data. No new participants were recruited and no approval number is asserted.

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CRediT authorship contribution statement

Ngo Nhat Quan: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review and editing. Pham Thai Son: Validation, Investigation, Visualization, Writing – review and editing.

Declaration of competing interests

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies

OpenAI Codex assisted with English-language editing, manuscript organisation, consistency checks against code and archived results, and reproducible plotting of existing aggregate data. It did not generate new experimental observations or replace the archived predictions. No other AI tool was used for this manuscript preparation. Both authors have read and approved the final manuscript and this disclosure; the authors remain responsible for the submitted content.

References

[1] R. S. Rosenberg and S. Van Hout, "The american academy of sleep medicine inter-scorer reliability program: sleep stage scoring," *Journal of Clinical Sleep Medicine*, vol. 9, no. 1, pp. 81–87, 2013.

[2] N. Jirakittayakorn, Y. Wongsawat, and S. Mitrirattanakul, "ZleepAnlystNet: a novel deep learning model for automatic sleep stage scoring based on single-channel raw EEG data using separating training," *Scientific Reports*, vol. 14, p. 9859, 2024.

[3] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," *arXiv preprint arXiv:1803.01271*, 2018.

[4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 770–778, 2016.

[5] A. Supratak, H. Dong, C. Wu, and Y. Guo, "DeepSleepNet: A model for automatic sleep stage scoring based on raw single-channel eeg," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 25, no. 11, pp. 1998–2008, 2017.

[6] E. Eldele, Z. Chen, C. Liu, M. Wu, C.-K. Kwoh, X. Li, and C. Guan, "An attention-based deep learning approach for sleep stage classification with single-channel eeg," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 29, pp. 809–818, 2021.

[7] H. Phan, K. Mikkelsen, O. Y. Chén, P. Koch, A. Mertins, and M. De Vos, "SleepTransformer: Automatic sleep staging with interpretability and uncertainty quantification," *IEEE Transactions on Biomedical Engineering*, vol. 69, no. 8, pp. 2456–2467, 2022.

[8] H. Phan and K. Mikkelsen, "Automatic sleep staging of EEG signals: recent development, challenges, and future directions," *Physiological Measurement*, vol. 43, no. 4, p. 04TR01, 2022.

[9] S. Haghayegh, K. Hu, K. Stone, S. Redline, and E. Schernhammer, "Automated sleep stages classification using convolutional neural network from raw and time-frequency electroencephalogram signals: Systematic evaluation study," *Journal of Medical Internet Research*, vol. 25, p. e40211, 2023.

[10] Z. He, M. Tang, P. Wang, L. Du, X. Chen, G. Cheng, and Z. Fang, "Cross-scenario automatic sleep stage classification using transfer learning and single-channel EEG," *Biomedical Signal Processing and Control*, vol. 81, p. 104501, 2023.

[11] E. Eldele, M. Ragab, Z. Chen, M. Wu, C.-K. Kwoh, X. Li, and C. Guan, "ADAST: Attentive cross-domain EEG-based sleep staging framework with iterative self-training," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 7, no. 1, pp. 210–221, 2023.

[12] J. Van Der Donckt, J. Van Der Donckt, E. Deprost, N. Vandenbussche, M. Rademaker, G. Vandewiele, and S. Van Hoecke, "Do not sleep on traditional machine learning: Simple and interpretable techniques are competitive to deep learning for sleep scoring," *Biomedical Signal Processing and Control*, vol. 81, p. 104429, 2023.

[13] B. Kemp, A. H. Zwinderman, B. Tuk, H. A. C. Kamphuisen, and J. J. L. Oberyé, "Analysis of a sleep-dependent neuronal feedback loop: the slow-wave microcontinuity of the eeg," *IEEE Transactions on Biomedical Engineering*, vol. 47, no. 9, pp. 1185–1194, 2000.

[14] PhysioNet, "Sleep-EDF database expanded, version 1.0.0." https://physionet.org/content/sleep-edfx/1.0.0/, 2013. Accessed 14 August 2026.

[15] S. F. Quan, B. V. Howard, C. Iber, J. P. Kiley, F. J. Nieto, G. T. O'Connor, D. M. Rapoport, S. Redline, J. Robbins, J. M. Samet, and P. W. Wahl, "The sleep heart health study: design, rationale, and methods," *Sleep*, vol. 20, no. 12, pp. 1077–1085, 1997.

[16] G.-Q. Zhang, L. Cui, R. Mueller, S. Tao, M. Kim, M. Rueschman, S. Mariani, D. Mobley, and S. Redline, "The national sleep research resource: towards a sleep data commons," *Journal of the American Medical Informatics Association*, vol. 25, no. 10, pp. 1351–1358, 2018.

[17] M. Massimini, R. Huber, F. Ferrarelli, S. Hill, and G. Tononi, "The sleep slow oscillation as a traveling wave," *Journal of Neuroscience*, vol. 24, no. 31, pp. 6862–6870, 2004.

[18] A. Rechtschaffen and A. Kales, *A Manual of Standardized Terminology, Techniques and Scoring System for Sleep Stages of Human Subjects*. Bethesda, MD: U.S. National Institute of Neurological Diseases and Blindness, 1968.

[19] R. B. Berry, R. Brooks, C. E. Gamaldo, S. M. Harding, R. M. Lloyd, S. F. Quan, M. T. Troester, and B. V. Vaughn, *The AASM Manual for the Scoring of Sleep and Associated Events: Rules, Terminology and Technical Specifications, Version 2.4*. Darien, IL: American Academy of Sleep Medicine, 2017.

[20] S. Davidson, R. Sharman, S. D. Kyle, and L. Tarassenko, “Is it time to revisit the scoring of slow wave (N3) sleep?,” *Sleep*, vol. 48, no. 10, p. zsaf063, 2025.

[21] M. Saerens, P. Latinne, and C. Decaestecker, “Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure,” *Neural Computation*, vol. 14, no. 1, pp. 21–41, 2002.

[22] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, pp. 1321–1330, 2017.

[23] F. Wilcoxon, “Individual comparisons by ranking methods,” *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.

[24] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. Chapman and Hall/CRC, 1994.

[25] S. Holm, “A simple sequentially rejective multiple test procedure,” *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.